

NEWTON, IA, USA

WMO: 725464

Lat: 41.674N

Lon: 93.022W

Elev: 953

StdP: 14.20

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04977

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.5	0.2	-13.2	2.8	-4.3	-8.4	3.6	1.0	30.2	16.3	26.8	20.4	8.0	320	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	91.0	77.4	89.6	76.9	86.3	74.5	80.6	88.1	78.5	85.8	76.7	84.1	10.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.9	155.3	86.6	76.7	144.1	84.3	73.5	128.9	81.1	45.3	88.6	42.8	85.1	40.9	84.7	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	22.4	19.6	-14.3	94.6	5.8	2.9	-18.5	96.7	-21.9	98.4	-25.1	100.1	-29.4	102.2
			-12.4	83.7	6.8	2.5	-17.3	85.5	-21.3	87.0	-25.2	88.4	-30.1	90.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	22.6	25.4	38.0	50.5	61.5	71.0	74.9	72.1	65.2	52.6	38.5	27.0
	DBStd	20.80	12.35	12.75	12.31	9.88	8.32	6.08	5.38	5.68	8.00	9.71	10.51	11.32
	HDD50	3234	849	691	408	111	9	0	0	0	2	85	366	713
	HDD65	6369	1314	1109	840	443	168	17	2	9	96	400	794	1177
	CDD50	3262	0	2	35	126	365	631	772	685	457	166	22	1
	CDD65	921	0	0	1	8	59	198	309	230	101	15	0	0
	CDH74	8411	0	0	18	135	574	1767	2935	1881	940	156	5	0
	CDH80	2856	0	0	1	31	162	599	1111	604	309	39	0	0
Wind	WSAvg	9.3	10.4	10.3	11.0	11.7	10.4	8.4	6.7	6.3	7.6	9.4	10.0	9.8
Precipitation	PrecAvg	36.3	0.9	1.2	2.0	3.8	4.8	5.6	4.7	4.4	4.1	2.3	1.9	1.1
	PrecMax	55.2	2.0	2.8	5.2	8.9	9.3	13.3	11.2	11.3	7.5	5.8	5.3	3.6
	PrecMin	24.8	0.4	0.2	0.0	1.4	0.7	1.0	0.6	0.6	1.8	0.5	0.0	0.0
	PrecStd	7.3	0.4	0.7	1.5	1.9	1.9	2.8	2.9	2.9	1.6	1.5	1.3	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	62.6	76.6	85.5	90.3	93.0	96.3	93.0	91.3	85.6	72.9	59.3
		MCWB	44.3	51.1	62.0	65.2	73.4	76.5	81.6	77.0	72.9	67.6	58.4	53.6
	2%	DB	47.7	54.4	69.2	77.4	84.2	89.8	91.2	89.8	87.9	78.7	64.4	52.1
		MCWB	41.8	47.4	57.8	61.1	69.3	76.4	79.5	77.6	73.1	64.0	54.4	46.7
	5%	DB	42.8	46.5	63.1	72.5	80.9	86.3	89.7	86.2	83.6	72.9	59.2	46.1
		MCWB	38.1	41.8	54.8	58.3	65.7	73.4	78.6	75.8	70.9	61.8	50.6	41.9
	10%	DB	37.2	42.6	56.6	68.0	76.6	83.6	86.1	82.4	80.6	68.3	54.4	41.4
		MCWB	34.5	38.7	48.4	55.6	64.4	72.1	76.5	73.4	68.2	59.5	47.5	37.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.9	53.3	64.4	68.0	76.7	81.6	84.7	82.9	78.3	71.1	60.7	55.2
		MCDB	52.4	60.9	72.6	79.1	85.9	89.5	93.6	89.9	87.4	79.1	68.9	59.5
	2%	WB	42.1	47.7	60.1	63.8	71.8	78.2	81.8	79.5	75.3	67.3	56.5	48.1
		MCDB	47.7	54.2	66.5	74.0	81.2	85.8	89.7	86.4	83.6	74.5	64.3	50.3
	5%	WB	38.3	42.7	54.7	60.3	68.9	76.0	79.6	77.2	72.9	64.2	52.1	42.1
		MCDB	42.3	47.3	61.2	70.0	76.9	83.9	86.6	84.2	80.5	70.0	58.2	46.3
	10%	WB	35.1	38.6	49.3	57.2	66.4	73.7	77.5	75.1	70.4	60.5	47.8	38.2
		MCDB	37.9	42.1	56.5	65.6	73.4	81.2	84.7	81.5	77.2	67.4	54.1	41.7
Mean Daily Temperature Range	5% DB	MDBR	15.9	16.2	18.2	21.0	19.6	18.9	18.8	18.5	22.1	20.1	17.7	15.0
		MCDBR	22.4	22.4	27.0	29.6	24.8	21.9	21.1	21.0	25.2	26.0	26.0	20.5
		MCWBR	17.2	16.2	17.2	16.8	12.6	12.3	13.0	12.7	13.3	14.5	16.9	16.1
	5% WB	MCDBR	21.3	21.5	24.5	25.1	21.0	19.9	19.0	18.4	21.2	21.4	23.7	19.2
MCWBR		16.6	16.4	17.1	16.5	12.7	13.4	13.5	12.8	12.9	14.6	17.5	15.8	
Clear-Sky Solar Irradiance	taub	0.288	0.310	0.331	0.380	0.409	0.420	0.437	0.417	0.385	0.341	0.300	0.288	
	taud	2.408	2.353	2.392	2.274	2.265	2.279	2.216	2.289	2.331	2.448	2.494	2.429	
	Ebn at Noon	274	284	291	282	275	272	265	268	269	270	269	264	
	Edh at Noon	25	31	34	41	42	42	44	40	36	28	23	23	
All-Sky Solar Radiation	RadAvg	606	927	1205	1502	1769	1962	1984	1722	1452	968	656	501	
	RadStd	58	65	115	115	133	191	141	108	107	113	56	45	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page